

Dry run day 5Q.1 Day 4 Recapa) $\text{down}(3) \rightarrow 321$

$\text{down}(3)$ print 3 $\rightarrow$ $\text{down}(2)$ print 2 $\rightarrow$ $\text{down}(1)$ print 1 $\rightarrow$ $\text{down}(0)$ return

b) $\text{up}(3) \rightarrow 123$

$\text{up}(3) \rightarrow \text{up}(2) \rightarrow \text{up}(1) \rightarrow \text{up}(0)$ return

then $\text{up}(1)$ print 1, $\text{up}(2)$ print 2, $\text{up}(3)$ print 3

c) $\text{both}(3) \rightarrow 321123$

print 3, 2, 1 $\rightarrow$ $\text{both}(0)$
return $\rightarrow$ print 1, 2, 3

d) $\text{skip}(5) \rightarrow 135$

$\text{skip}(5) \rightarrow \text{skip}(3) \rightarrow \text{skip}(1) \rightarrow \text{skip}(-1)$ return
($n \leq 0$)
print 1, 3, 5

e) digitSum (1234) $\rightarrow$ 10

$$ds(1234) = 4 + ds(123)$$

$$ds(123) = 3 + ds(12)$$

$$ds(12) = 2 + ds(1)$$

$$ds(1) = 1 + ds(0) = 1 + 0 = 1$$

$$2 + 1 = 3 \rightarrow 3 + 3 = 6 \rightarrow 4 + 6 = 10$$

f) noBase(5) $\rightarrow$

$$5 + noBase(4) \rightarrow 4 + noBase(3) \rightarrow \dots$$

$$\rightarrow 0 \rightarrow -1 \rightarrow -2 \dots$$

In catch error will print because there is no base case

g) power(2,5) $\rightarrow$ 32 in 6 calls

$$p(2,5) \rightarrow p(2,4) \rightarrow p(2,3) \rightarrow p(2,2) \rightarrow p(2,1) \rightarrow p(2,0)$$

$$\text{return } p(2,0) = 1, p(2,1) = 2, p(2,2) = 4, p(2,3) = 8$$

$$p(2,4) = 16, p(2,5) = 32$$

h) fact(4)

$$fact(4) \rightarrow fact(3) \rightarrow fact(2) \rightarrow fact(1)$$

fact(1) returns 1

$$fact(2) \quad r = 2 * 1 = 2$$

returns 2

$$fact(3) \quad r = 3 * 2 = 6$$

returns 6
 $\text{fact}(4) = 4 \times 6 = 24$
 returns 24
 main print 24

i) mystery
 $\text{mystery}(12) = 0 + 10^4 \times \text{mystery}(6)$
 $\text{mystery}(6) = 0 + 10^4 \times \text{mystery}(3)$
 $\text{mystery}(3) = 0 + 10^4 \times \text{mystery}(1)$
 $\text{mystery}(1) = 0 + 10^4 \times \text{mystery}(0) = 1$
~~mystery~~ $m(3) = 110211 \rightarrow m(6) = 0110211$
~~mystery~~ $\rightarrow m(12) = 0110021100$

$\text{mystery}(5) = 1 + 10^4 \times \text{mystery}(2)$
 $\text{mystery}(2) = 0 + 10^4 \times \text{mystery}(1) = 10$
~~mystery~~ $\& m(5) = 1 + 10^8 = 101$

j) rev (java)

$\text{rev}(\text{java}) = \text{rev}(\text{ava}) + j$
 $\text{rev}(\text{ava}) = \text{rev}(\text{va}) + a$
 $\text{rev}(\text{va}) = \text{rev}(\text{a}) + v$
 $\text{rev}(\text{a}) = a$
 $(\text{length} \leq 1)$
 $a + v = av \rightarrow av + a = ava \rightarrow ava + j = avaj$

k) isEven / isOdd

isEven(4) $\rightarrow$ isOdd(3) $\rightarrow$ isEven(2) $\rightarrow$
isOdd(1) $\rightarrow$ isEven(0) = true
isOdd(3) $\rightarrow$ isEven(2) $\rightarrow$ isOdd(1) $\rightarrow$ isEven
(0) = true

isEven(3) $\rightarrow$ isOdd(2) $\rightarrow$ isEven(1) $\rightarrow$ isOdd(0)
= false

l) fib(5) $\rightarrow$ fib(5) = 5 calls = 15

$$\text{calls}(n) = 1 + \text{calls}(n-1) + \text{calls}(n-2)$$

n	0	1	2	3	4	5
fib(n)	0	1	1	2	3	5
calls	1	1	3	5	9	15

m) hamdi: $\text{moves}(n) = 2 \times \text{moves}(n-1) + 1 = 2^n - 1$

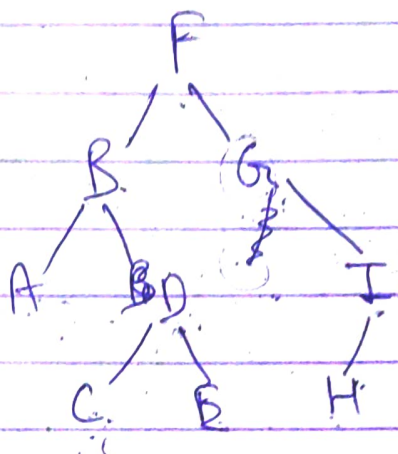
n	moves
3	7
10	1023
20	1048575

n) linked list 1 $\rightarrow$ 2 $\rightarrow$ 3

$\text{printReverse}(1) \rightarrow \text{printReverse}(2) \rightarrow \text{printReverse}(3) \rightarrow \text{printReverse}(\text{null}) \text{ return}$

$\text{print } 3, \text{ print } 2, \text{ print } 1 \rightarrow 3 \ 2 \ 1$
 $\text{length}(1) = 1 + \text{length}(2) = 1 + 1 + \text{length}(3) = 1 + 1 + 1 + \text{length}(\text{null}) = 0 \rightarrow 3$
 O/P O/P 3 2 1 length = 3

Q.2 Binary Tree Basics



Preorder (Root, Left, Right) $\rightarrow F \ B \ A \ D \ C \ E \ G \ I \ H$
 $F \rightarrow B \rightarrow A \text{ (Leaf)} \rightarrow D \rightarrow C \text{ (leaf)} \rightarrow E \text{ (leaf)} \rightarrow G \text{ (leaf)}$
 $\rightarrow G \rightarrow (G.\text{left null}) \rightarrow I \rightarrow H$

Inorder (Left, Root, Right) $\rightarrow A \ B \ C \ D \ E \ F \ G \ H \ I$
 (sorted)

Postorder (Left, Right, Root) $\rightarrow A \ C \ E \ D \ B \ H \ I \ G \ F$

Level order (queue) $\rightarrow$ F B G A D I C E H

Step	poll	print	queue
Start	-	-	[F]
1	F	F	[B, G]
2	B	B	[G, A, D]
3	G	G	[A, D, I]
4	A	A	[D, I]
5	D	D	[I, C, E]
6	I	I	[C, E, H]
7	C	C	[E, H]
8	E	E	[H]
9	H	H	[]

printLevels - count = q.size()

level 0 : F

level 1 : B G

level 2 : A D I

level 3 : C E H

Size/height/ leaves

size(H) = 1 size(I) = 1 + size(H) + 0 = 2, size(G) = 1 + 0 + 2 = 3

size(C) = size(E) = 1, size(D) = 3, size(A) = 1, size(B) = 1 + 1 + 3 = 5

size(P) = 1 + 5 + 3 = 9

Level order (queue) $\rightarrow$ F B G A D I C E H

Step	poll	print	queue
Start	-	-	[F]
1	F	F	[B, G]
2	B	B	[G, A, D]
3	G	G	[A, D, I]
4	A	A	[D, I]
5	D	D	[I, C, E]
6	I	I	[C, E, H]
7	C	C	[E, H]
8	E	E	[H]
9	H	H	[]

printLevels - count = q.size()

level 0 : F

level 1 : B G

level 2 : A D I

level 3 : C E H

Size/height/ leaves

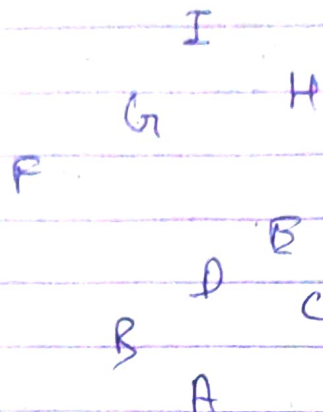
size(H) = 1 size(I) = 1 + size(H) + 0 = 2, size(G) = 1 + 0 + 2 = 3

size(C) = size(E) = 1, size(D) = 3, size(A) = 1, size(B) = 1 + 1 + 3 = 5

size(F) = 1 + 5 + 3 = 9

$\text{height}(H) = 0$ $I = 1$ $G = 2$; $A, C, E = 0$, $D = 1$
 $\text{height}(F) = 1 + \max(2, 2) = 3$ (empty tree $= -1$)
 Leaves : $A, C, E, H = 4$

print level ways



Q.3 Array Binary Tree.java

$\text{left}(i) = 2i+1$, $\text{right}(i) = 2i+2$ $\text{parent}(i) = (i-1)/2$
 Index $>=$ size null child

Array A B C D E F G H I (size=9)

i	node	parent	left(2i+1)	right(2i+2)
0	A	-	1 → B	2 → C
1	B	A	3 → D	4 → E
2	C	A	5 → F	6 → G
3	D	B	7 → H	8 → I
4	E	B	9 (≥ 9) → -	10 → -
5	F	C	11 → -	12 → -
6	G	C	13 → -	14 → -
7	H	D	15 → -	16 → -
8	I	D	17 → -	18 → -

Preorder A B D H I E C F G

pre(0) A → pre(1) B → pre(3) D → pre(7) H →
 pre(15) out of range return
 → pre(16) return → back D: pre(8) I → return
 → back B: pre(4) E → pre(9), pre(10) return
 → back A: pre(2) C → pre(5) F → pre(6) G

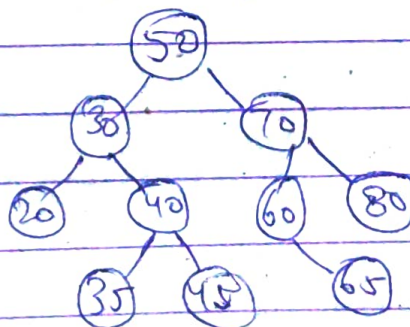
Inorder → H D I B E A C G

in(0) → in(1) → in(3) → in(7): left out, print H
 , right out
 → print D → in(8): print I → print B → in(4)
 : print E
 → print A → in(2) → in(5): print F → print C
 → in(6): print G

Height - $\text{height} = 31 - \text{numberOfLeadingZeros}(9)$
 9 binary = 1001 (4 bit), leading zeros = 28
 Height = $31 - 28 = 3$ (= floor(log 29))

④ BST.java

50 30 70 20 40 60 80 35 45 65



Inorder : 20 30 35 40 45 50 60 65 70
 Preorder : 50 30 20 40 35 45 70 60 65 80
 Postorder : 20 35 45 40 30 65 60 80 70 50
 Level order : 50 30 20 40 60 80 35 45 65

min / max / search min : 50 → 30 → 20
 min → 20

max : 50 → 70 → 80
 → 80

Height = 3

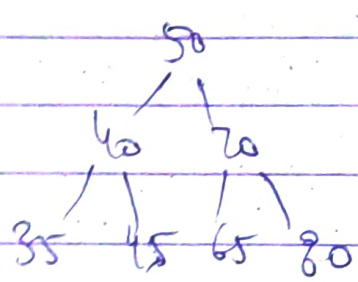
contains (65) : 50 → 70 → 60 → 65 → true →

contains (66) : false

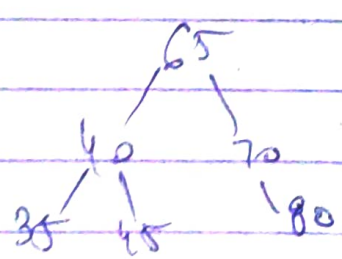
Delete 20 inorder : 30 35 40 45 50 60 65 70 80

delete 60 70.left = 65

inorder : 30 35 40 45 50 65 70 80



delete 50

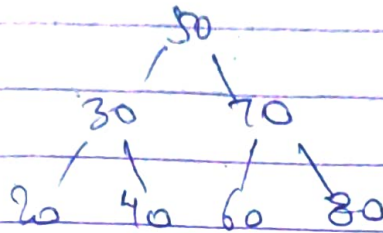


Complexity $\log n \rightarrow O(\log n)$

sorted insert $O(n)$

⑤ BST Lab

1 → 7 → 50 → 30 20 20 40 60 80



DFS push 50

stack: [50]

pop 50, print

push 70, push 30

stack [30, 70]

top = 30

pop 30 print

push 40 push 20

stack [20, 40, 70]

pop 20 print

(leaf)

pop 40 print

(leaf)

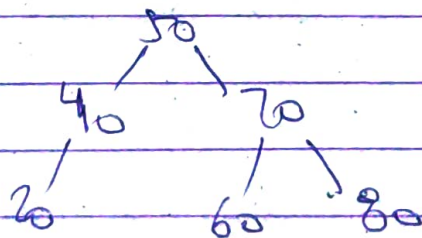
pop 70 print

push 80, push 60

stack [60, 80]

pop 60 print

pop 80 print



⑥ Iterative Traversals

Preorder 50 30 20 40 70 60 80

pop 50 print $\rightarrow$ push 70, 30 stack 30 70

pop 30 print $\rightarrow$ push 40, 20 stack 20 40 70

pop 20 print $\rightarrow$ pop 40 stack 70

pop 70 print $\rightarrow$ push 80 60, stack 60, 80

pop 60 print $\rightarrow$ pop 80 stack empty

Inorder 20 30 40 50 60 70 80

push = 50 30 20

stack 20 30 50

pop 20 print (right null)

pop 30 print right 40: push, pop 40 print

pop 50 print $\rightarrow$ right 70: push 70, 60

pop 60 print, pop 70 print $\rightarrow$ right 80: push, pop 80 print

Postorder 20 40 30 60 80 70 50

s1 = [50]

pop 50

pop 70

pop 80

pop 60

pop 30

pop 40

push 30 70

push 60 80

push 20, 40

pop 20 $\rightarrow$ s2: 20

s2 ~~the~~ order = 50, 70, 80 60 30 40 20
(root, right, left)

82 pop = 20 40 30 60 80 70 50

(7) Java Tree Demo

input = 72, 45, 91, 60, 88, 33, 79

print set	[33, 45, 60, 72, 79, 88, 91]
first / last	33 / 91
floor (80)	79
ceiling (80)	88
headset (60)	[33, 45]
descending set	[91, 88, 79, 72, 60, 45, 33]

TreeMap = [Arjun, Kabeer, Meera, Zoya] first Arjun = 3
last Zoya = 27

PriorityQueue :-

index $n/2 - 1 = 2$ to 0 shiftDown

[72, 45, 91, 60, 88, 33, 79] $i=2$: 91 vs children

33, 79 $\rightarrow$ 33

swap [72, 45, 33, 60, 88, 91, 79]

$i=1$ 45 vs 60, 88 $\rightarrow$ no swap

$i=0$ 72 vs 45 33 $\rightarrow$ 33 swap

[33, 45, 72, 60, 88, 91, 79]

poll() = 33 45 60 72 79 88 91

⑧ Tree Types Check

① Perfect 4(2(1,3), 6(5,7))
 full T | complete T (7 nodes, index < 7)
 perfect T (size $7 = 2^3 - 1$) | balanced T | BST T

② complete 4(2(1,3), 6(5,-))

full F (left of 6) | complete T (6 nodes, index 0...5)
 perfect F (size 6, 7 wanted) | balanced T (6: left 6, right -1) | BST T

③ full not complete 1(2, 3(4,5))

full T | complete F (4 index 5 $\geq$ size 5) perfect F (size 5) | balanced T (left 0, right 1) | BST F (2 is in left 2 $>$ 1)

④ degenerate 1 $\rightarrow$ 2 $\rightarrow$ 3 $\rightarrow$ 4

full F (1's left right) | complete F (3's index 6 $\geq$ 4) | perfect F (size 4, 15 wanted) | balanced F (root: left -1, right 2) | BST T (1 $<$ 2 $<$ 3 $<$ 4)

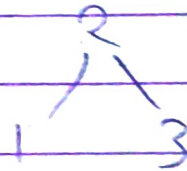
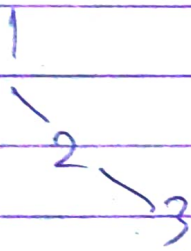
⑤ BST 10(5(-12, 15))

full F (1's right) | complete F (3's index
 $6 \geq 4$) perfect F (size 4, 15 wanted) |
 balanced F (root: left = 1, right = 2)

full F (5 right) | complete F (12's index
 $4 \geq 4$)
 perfect F | balanced T
 BST F (12 range (5, 10), $12 \geq 10$)

9> Degenerate BST

sorted insert 1, 2, 3:
 insert 2, 1, 3



$$\text{height} = 2(n-1) = 1$$

$$n = 20,000$$

Height	37
Steps to find 19,999	13
insert	$n \log n$
$\log_2(20000) = 14.3$	